```dockerfile
FROM node:20-alpine AS builder
WORKDIR /build
COPY package*.json ./
RUN npm ci && npm cache clean --force
COPY . .
RUN npm run lint

FROM node:20-alpine AS production
RUN addgroup -g 1001 -S nodejs && adduser -S nodeuser -u 1001 -G nodejs
WORKDIR /app
COPY --from=builder --chown=nodeuser:nodejs /build/node_modules ./node_modules
COPY --chown=nodeuser:nodejs server.js package.json ./
USER nodeuser
EXPOSE 3000
HEALTHCHECK --interval=30s --timeout=5s --start-period=20s --retries=3 CMD node -e "require('http').get('http://localhost:3000/heal
CMD ["node", "server.js"]
```

```
PS C:\Users\capta\kitchen-app> Invoke-RestMethod http://localhost:3000/health

PS C:\Users\capta\kitchen-app> kubectl get all
NAME                                    READY   STATUS             RESTARTS        AGE
pod/kitchen-api-5947bc6445-9qxfs        0/1     CrashLoopBackOff   6 (3m17s ago)   8d
pod/kitchen-db-96d65f798-48hbs          1/1     Running            1 (11m ago)     8d
pod/kitchen-web-ui-674df5754d-wbg9n     1/1     Running            0               8d

NAME                    TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
service/kitchen-api     ClusterIP   10.104.222.251   <none>        3000/TCP       8d
service/kitchen-db      ClusterIP   10.108.59.126    <none>        27017/TCP      8d
service/kitchen-web-ui  NodePort    10.103.40.195    <none>        80:30007/TCP   8d
service/kubernetes      ClusterIP   10.96.0.1        <none>        443/TCP        8d
```

```dockerfile
FROM node:20-alpine AS builder
WORKDIR /build
COPY package*.json ./
RUN npm ci && npm cache clean --force
COPY . .
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FROM node:20-alpine AS production
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USER nodeuser
EXPOSE 3000
HEALTHCHECK --interval=30s --timeout=5s --start-period=20s --retries=3 CMD node -e "require('http').get('http://localhost:3000/heal
CMD ["node", "server.js"]
```

```
PS C:\Users\capta\kitchen-app> docker build -t kitchen-web:local ./web-ui
 => [1/2] FROM docker.io/library/nginx:alpine@sha256:f46cb72c7df02710e693e863a983ac42f6a9579058a59a35f1ae36c9958e4ce0                0.0s
 => => resolve docker.io/library/nginx:alpine@sha256:f46cb72c7df02710e693e863a983ac42f6a9579058a59a35f1ae36c9958e4ce0                0.0s
 => CACHED [2/2] COPY . /usr/share/nginx/html                                                                                        0.0s
 => exporting to image                                                                                                               0.1s
 => => exporting layers                                                                                                              0.0s
 => => exporting manifest sha256:299079976c2fad6f66aed8d3819d6255e790e5a7e2d0620165cf899c59ae70d                                     0.0s
 => => exporting config sha256:2b36189217d14f5c7d1d4d9200f4357ee190341a1f0873c7ecb5418c5bf4c7f9                                      0.0s
 => => exporting attestation manifest sha256:9a3222fbcbc56e7284e209393f0fe768149f2c81419a5d14bc4c3daa9d67dbe7                        0.0s
 => => exporting manifest list sha256:8b33c2a09c8cca16d5c8a634e86edf1738c1ad6e87deca91c3b0077b14d26273                               0.0s
 => => naming to docker.io/library/kitchen-web:local                                                                                 0.0s
 => => unpacking to docker.io/library/kitchen-web:local                                                                              0.0s
PS C:\Users\capta\kitchen-app>
```

```
PS C:\Users\capta\kitchen-app> docker build -t kitchen-web:local ./web-ui
 => => exporting manifest list sha256:8b33c2a09c8cca16d5c8a634e86edf1738c1ad6e87deca91c3b0077
 => => naming to docker.io/library/kitchen-web:local
 => => unpacking to docker.io/library/kitchen-web:local
PS C:\Users\capta\kitchen-app> cd k8s
PS C:\Users\capta\kitchen-app\k8s> kubectl apply -f .
deployment.apps/kitchen-api unchanged
service/kitchen-api unchanged
deployment.apps/kitchen-db unchanged
service/kitchen-db unchanged
deployment.apps/kitchen-web-ui unchanged
service/kitchen-web-ui unchanged
PS C:\Users\capta\kitchen-app\k8s>
```

```
PS C:\Users\capta\kitchen-app\k8s> kubectl apply -f .
deployment.apps/kitchen-api unchanged
service/kitchen-api unchanged
deployment.apps/kitchen-db unchanged
service/kitchen-db unchanged
deployment.apps/kitchen-web-ui unchanged
service/kitchen-web-ui unchanged
PS C:\Users\capta\kitchen-app\k8s> kubectl get pods
NAME                              READY   STATUS              RESTARTS        AGE
kitchen-api-5947bc6445-9qxfs      0/1     RunContainerError   3 (34s ago)     8d
kitchen-db-96d65f798-48hbs        1/1     Running             1 (3m47s ago)   8d
kitchen-web-ui-674df5754d-wbg9n   1/1     Running             0               8d
PS C:\Users\capta\kitchen-app\k8s>
```

127.0.0.1:64918 Paused

Kitchen App Running

Web UI is working.

Windows PowerShell

```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\capta> & "C:\Program Files\Kubernetes\Minikube\minikube.exe" start
* minikube v1.38.1 on Microsoft Windows 11 Home Single Language 25H2
* Using the docker driver based on existing profile
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.50 ...
* Restarting existing docker container for "minikube" ...
* Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
* Verifying Kubernetes components...
  - Using image gcr.io/k8s-minikube/storage-provisioner:v5
* Enabled addons: storage-provisioner, default-storageclass
* Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
PS C:\Users\capta> & "C:\Program Files\Kubernetes\Minikube\minikube.exe" image load kitchen-api:local
PS C:\Users\capta> & "C:\Program Files\Kubernetes\Minikube\minikube.exe" image load kitchen-web:local
PS C:\Users\capta> & "C:\Program Files\Kubernetes\Minikube\minikube.exe" service kitchen-web-ui
```

NAMESPACE	NAME	TARGET PORT	URL
default	kitchen-web-ui	80	http://192.168.49.2:30007

```
* Starting tunnel for service kitchen-web-ui.
```

NAMESPACE	NAME	TARGET PORT	URL
default	kitchen-web-ui		http://127.0.0.1:64918

```
* Opening service default/kitchen-web-ui in default browser...
! Because you are using a Docker driver on windows, the terminal needs to be open to run it.
```

File Edit Selection View Go Run Terminal Help — kitchen-app

EXPLORER

- OUTLINE
- KITCHEN-APP
  - api
    - node_modules
    - .eslintrc.json U
    - Dockerfile U
    - package-lock.json U
    - package.json U
    - server.js U
    - server.test.js U
  - docs
  - k8s
    - api.yaml U
    - mongodb.yaml U
    - web-ui.yaml U
  - web-ui
    - Dockerfile U
    - index.html U
  - docker-compose.yml U

package.json U | Dockerfile api U | server.js U | server.test.js U | docker-compose.yml U | index.html U | Dockerfile web-ui U

api > Dockerfile > ...

```
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WORKDIR /build
COPY package*.json ./
RUN npm ci && npm cache clean --force
COPY . .
RUN npm run lint

FROM node:20-alpine AS production
RUN addgroup -g 1001 -S nodejs && adduser -S nodeuser -u 1001 -G nodejs
WORKDIR /app
COPY --from=builder --chown=nodeuser:nodejs /build/node_modules ./node_modules
COPY --chown=nodeuser:nodejs server.js package.json ./
USER nodeuser
EXPOSE 3000
HEALTHCHECK --interval=30s --timeout=5s --start-period=20s --retries=3 CMD node -e "require('http').get('http://localhost:3000/h
CMD ["node", "server.js"]
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS — powershell

```
PS C:\Users\capta\kitchen-app> kubectl get nodes -o wide
.2.1
PS C:\Users\capta\kitchen-app> kubectl get svc kitchen-web-ui
NAME          TYPE       CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
kitchen-web-ui   NodePort   10.103.40.195   <none>        80:30007/TCP   8d
PS C:\Users\capta\kitchen-app> Invoke-RestMethod http://localhost:3000/health

status db
------ --
ok     connected

PS C:\Users\capta\kitchen-app>
```

TIMELINE

127.0.0.1:64918

Kitchen App Running

Web UI is working.

File Edit Selection View Go Run Terminal Help

kitchen-app

EXPLORER

OUTLINE

KITCHEN-APP

- api
  - node_modules
  - .eslintrc.json U
  - Dockerfile U
  - package-lock.json U
  - package.json U
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  - server.test.js U
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  - api.yaml U
  - mongodb.yaml U
  - web-ui.yaml U
- web-ui
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  - index.html U
- docker-compose.yml U

TIMELINE

package.json U | Dockerfile api U | server.js U | server.test.js U | docker-compose.yml U | index.html U | Dockerfile web-ui U

api > Dockerfile > ...

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FROM node:20-alpine AS builder
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PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

powershell

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PS C:\Users\capta\kitchen-app> kubectl get all
NAME                                  READY   STATUS             RESTARTS        AGE
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pod/kitchen-web-ui-674df5754d-wbg9n   1/1     Running            0               8d

NAME                  TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
service/kitchen-api   ClusterIP   10.104.222.251   <none>        3000/TCP       8d
service/kitchen-db    ClusterIP   10.108.59.126    <none>        27017/TCP      8d
service/kitchen-web-ui NodePort   10.103.40.195    <none>        80:30007/TCP   8d
service/kubernetes    ClusterIP   10.96.0.1        <none>        443/TCP        8d
```